



What happens if the world runs out of sand?

【BBC精选：沙子用完怎么办？ | 沙漠里的沙子不能用来建房子】

Summary: In 2008, a Jamaican beach disappeared overnight with 500 truckloads of sand stolen. This incident reveals the global sand theft crisis driven by high demand. Sand forms over millennia, carrying Earth's history through color and composition. It is vital for concrete, glass, and even silicon chips. However, desert sand is unusable for construction, leading to ecological damage from river and coastal mining. Sustainable practices are urgently needed to protect this precious resource.

摘要：2008年牙买加一处海滩的沙子一夜被盗，揭开了全球沙粒盗窃危机的序幕。沙粒形成需数千年，其颜色和成分记录着地球历史。它是制造混凝土、玻璃乃至硅芯片的关键原料。但沙漠沙因过于光滑无法用于建筑，导致河流与海岸采沙加剧，引发生态灾难。专家呼吁采用可持续方案保护这一濒危资源。





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双语字幕 | 生词高亮 | 智能文稿

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▶ In 2008, an entire beach and Jamaica vanished.

2008年，一整片海滩和牙买加消失了。

▶ Five hundred truckloads of glittering white sand was stolen under the cover of darkness and never found.

五百卡车闪闪发光的白色沙子#在夜幕掩护下#被偷走，从此下落不明。

▶ In fact, sand is being stolen all over the world.

事实上，全球各地都在发生沙子盗窃事件。

▶ Highlighting how this seemingly plentiful material is now so valuable

这凸显出这种#看似##取之不尽#的材料如今变得如此珍贵

▶ that people will go to extraordinary lengths to obtain it.

以至于人们会不惜采取极端手段来获取它。

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▶ Sand isn't just the gritty granules that trap between your toes when you're at the beach.

沙子不仅仅是海滩上陷进脚趾间的粗颗粒。

▶ It's a mirror of the earth's history that takes thousands if not millions of years to form.

它是地球历史的镜像，需要数千甚至数百万年才能形成。

▶ The colours of sand tell stories of landscapes long gone.

沙子的色彩诉说着早已消失的地貌故事。

▶ On tropical beaches, sand gleams white, formed from the crushed skeletons of coral and marine life.

热带海滩上，沙子洁白闪亮，由#珊瑚#和海洋生物的碎骨形成。

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▶ Along volcanic shores it turns black, from basalt-born of molten lava.

火山海岸边，它因源自#熔融##岩浆#的玄武岩而变成黑色。

▶ Red sands carry the tale of ancient iron-rich rocks oxidised over time.

红色沙粒承载着富铁岩石经年氧化的#传说#。

▶ And it's not just what sands made of, it's also what can live in it.

不仅在于沙子的构成，还在于其中能栖息的生物。

▶ The Macher sand dunes in Scotland are home to hundreds of plants and animals and show just how vibrant and full of life sand can be.

苏格兰的马赫尔沙丘是数百种动植物的家园，展现出沙子所能孕育的蓬勃生机。

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▶ Like tree rings, the layers of a sand dune tell its history.

如同年轮，沙#丘#的层理诉说着它的历史。

▶ In the Narmib desert, scientists use optical luminescence to reveal the sand's age,

在纳米布沙漠，科学家利用#光学#释光技术探测沙粒年龄，

▶ calculating the last time it saw sunlight.

#测算#其最后一次接触#阳光#的时间。

▶ In Utah, petrified Jurassic dunes formed 180 million years ago are hardened into a magnificent alien landscape.

犹他州，一亿八千万年前形成的侏罗纪沙丘已石化成#壮丽#而#奇异#的景观。

▶ Sand exhibits mysterious and unusual physics.

沙子展现出神秘非凡的物理特性。

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▶ In the Sahara, singing sand dunes produce a haunting home heard for kilometres.

撒哈拉沙漠中，鸣响的沙丘能发出萦绕数公里的空灵低鸣。

▶ This phenomenon occurs when a disturbance like a small avalanche sets the dune vibrating like a plucked cello string.

当小型崩塌等扰动使沙丘如大提琴弦般振动时，便会产生这种现象。

▶ 28 million years ago a meteor struck the Libyan desert sand

2800万年前，一颗#陨石#撞击利比亚沙漠沙地，

▶ and out of the intense forces forged one of the rarest materials.

在#剧烈#力量作用下锻造出最稀有的物质之一。

▶ Libyan desert glass.

利比亚沙漠玻璃。

▶ So rare and prized, it sits in the centre of Tutankhamans breastplate carved as a golden scarab.

因其稀世珍贵，被雕成金甲虫镶嵌于图坦卡蒙胸甲中央。

▶ Sand's versatility as a building material was understood by the Romans.

罗马人早已洞悉沙子作为建筑材料的#多功能性#。

▶ They figured out how to make sand with lime and volcanic ash to create concrete.

他们发现将沙子与#石灰#、火山#灰#混合可制成混凝土。

▶ An invention that's literally stood the test of time.

这项发明真正经受住了时间的考验。

▶ The Romans constructed fast networks of roads, aqueducts and structures whose remains have survived for two millennia.

罗马人建造的道路、引水渠和建筑网络留存两千年至今。

▶ The Pantheon in Rome has the world's largest unreinforced concrete dome.

罗马万神殿拥有世界上最大的无筋混凝土#穹顶#。

▶ Four thousand years ago in ancient Mesopotamia,

四千年前的古美索不达米亚，

▶ humans first crafted glass using a simple mixture of sand, lime and soda.

人类首次用沙子、#石灰#与#苏打#的简单混合物制作玻璃。

▶ Heated then rapidly cooled.

加热后急速冷却。

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▶ Today it's in the windows we peer through the screens we touch

如今它存在于我们#凝视#的窗扉、触摸的屏幕，

▶ and the fiber optic cables that carry our thoughts close to the speed of light.

以及以近光速传递思想的#光纤#电缆中。

▶ Silicon chips, the brains behind everything from smartphones to space shuttles are also made from precious quartz sand.

从智能手机到航天飞机，其核心#硅#芯片同样源自#珍贵#的#石英#砂。

▶ We're literally surfing the internet on sand particles.

我们简直是在沙粒上遨游互联网。

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▶ It's hard to imagine our world running out of sand when we have deserts like the Sahara covering 8% of the planet and totaling over 9 million square kilometres.

当撒哈拉等沙漠覆盖地球8%面积、总面积超900万平方公里时，很难想象我们会面临沙荒。

▶ But desert sand, with its smooth rounded grains shaped by wind, is useless in construction.

但经风蚀形成光滑圆粒的沙漠沙在建筑中毫无用处。

▶ It's like trying to build with marbles.

犹如试图用弹珠建造。

▶ Instead, the world's insatiable need for concrete buildings, bridges and motorways

相反，世界对混凝土建筑、桥梁和高速公路#永不满足#的需求，

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▶ demands sand from rivers and coastlines which has angular grains that lock together like a jigsaw puzzle.

要求采掘河流与海岸的棱角砂粒——它们能如#拼图#般紧密咬合。

▶ And this usable sand is running out.

而这类可用沙正在枯竭。

▶ Mining sand from rivers and coastlines disrupts ecosystems and threaten species that depend on those sediments.

开采河沙与海沙会破坏生态系统，威胁依赖这些沉积物的物种。

▶ Sand flows down rivers protecting shorelines, builds deltas and helps prevent flooding.

顺流而下的沙粒保护海岸线、构筑三角洲、助力防洪。

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▶ When too much is taken, a chain reaction is set off leading to severe floods and erosion, 过量开采会引发连锁反应，导致严重洪涝与#侵蚀#，

▶ making us more vulnerable to nature's forces and scarcity leads to crime.

让我们更#易受#自然之力侵袭，而#稀缺性#则滋生犯罪。

▶ And parts of the world so call sand mafias, illegal mining gangs.

世界某些地区所谓的"沙黑帮"——非法采砂团伙，

▶ Dredge for sand so intensely that it's causing riverbanks to collapse, destroying villages.

#疯狂## dredge#采砂导致河岸坍塌、村庄毁灭。

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▶ Those who oppose that often face violent retaliation.

反对者常遭遇暴力报复。

▶ Sand has become a commodity so valuable that people are willing to kill for it.

沙子已成为值得为之杀戮的珍贵#商品#。

▶ Today, concrete is the most common building material on the planet.

如今混凝土是地球上最普遍的建筑材料。

▶ If it were a country, it would rank among the world's top carbon emitters.

若将其视作国家，其碳排放量将位列世界前三。

▶ If we were to embrace more sustainable building practices based on local materials and conditions,

若能基于当地材料与条件#采用#更#可持续#的建筑方式，

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▶ sand could be preserved for essential purposes and protected from over extraction.

沙子便可留作 essential 用途，避免过度#开采#。

▶ Vulnerable rivers and dunes could also be protected using cattle or sheep to naturally manage vegetation,

可用#牛#羊自然管理#植被#以保护#脆弱#河流与沙丘，

▶ shift sand and boost biodiversity, or by planting natural barriers like mangroves to shield coastlines.

迁移沙粒、#增强##生物多样性#，或种植红树林等天然屏障#防护#海岸。

▶ So the next time you're walking on a beach, take a moment to feel the sand under your feet.

当下次漫步海滩时，请驻足感受脚下的沙粒。

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▶ You're standing on one of humanity's most prized resources.

你正立于人类最珍贵的资源之上。

▶ The remains of ancient ecosystems, and perhaps even the stuff that powers your smartphone.

那是远古生态系统的遗存，或许更是驱动你智能手机的源泉。

▶ These tiny grains of sand are our history, sure, but they're also our future.

这些微小沙粒是我们的历史，但无疑也是我们的未来。

